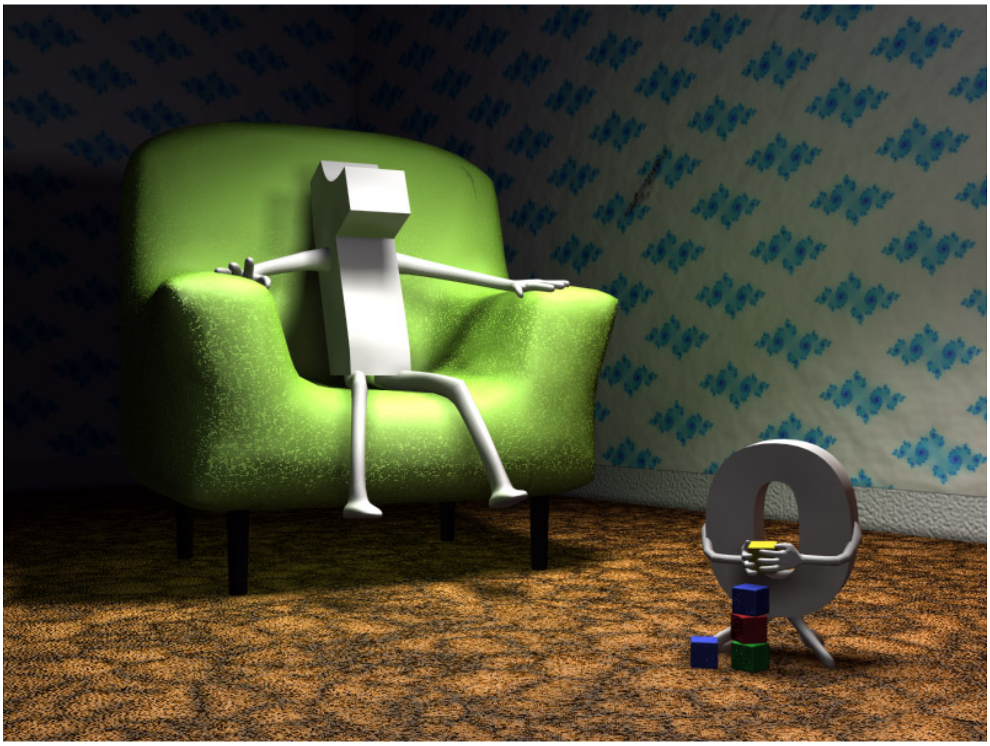
Quiz Week 1

Integers



Divisibility. Gcd. Euclidean Algorithm.

1. **In each case, determine whether the statement is true or false.**
   (1) Suppose that n and m are two integers not both zero, and $d > 1$ is an integer satisfying that $d = xm + yn$ for some integers x and y. Then $d = \gcd(n, m)$.
   (2) If $a|b$ and $c|d$, then $ac|bd$.
   (3) -175 and 72 are coprime. If true, use the Euclidean Algorithm to express 1 as a linear combination of -175 and 72.

Solution: (1) **False:** Take $n = 4$, $m = 8$ and $d = 12$. Notice that 12 satisfies $12 = 1 \cdot 4 + 1 \cdot 8$ but $\gcd(4, 8) = 4 \neq 12$.

(2) **True:** From $a|b$ we have that $b = an$ for some $n \in \mathbb{Z}$. Similarly, from $c|d$ we have that $d = cm$ for some $m \in \mathbb{Z}$. But then

$$bd = (an)(cm) = (ac)(nm),$$

which implies that $ac|bd$.

(3) **True:** We use the Euclidean Algorithm to compute $\gcd(-175, 72)$.

$$\begin{aligned}
-175 &= (-3) \cdot 72 + 41 \\
72 &= 1 \cdot 41 + 31 \\
41 &= 1 \cdot 31 + 10 \\
31 &= 3 \cdot 10 + 1 \\
10 &= 1 \cdot 10 + 0.
\end{aligned}$$

Since the last remainder before the zero remainder is 1, we can conclude that -175 and 72 are coprime, i.e., $\gcd(-175,\ 72) = 1$. To finish, it remains to express 1 as a linear combination of -175 and 72.

$$\begin{aligned}
1 &= 31 - 3 \cdot 10 = 31 - 3(41 - 31) = (-3)41 + 4 \cdot 31 = \\
&= (-3)41 + 4(72 - 41) = 4 \cdot 72 + (-7)41 = 4 \cdot 72 + (-7)(-175 + 3 \cdot 72) = \\
&= (-17) \cdot 72 + (-7) \cdot (-175)
\end{aligned}$$

Thus: $\gcd(-175,\ 72) = 1 = (-7) \cdot (-175) + (-17) \cdot 72$.